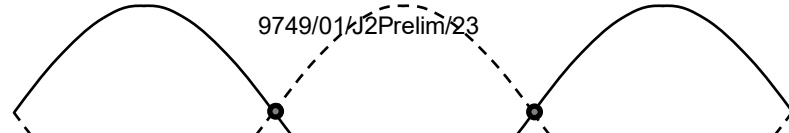


- 17** The stationary wave shown below is the result of the superposition of two identical waves travelling in opposite directions.

NYJC 2023



[Turn over

Which of the following best states the phase difference between the two waves at points X, Y and Z?

	X	Y	Z
A	$-\pi$	0	π
B	$-\pi/2$	0	$\pi/2$
C	$\pi/2$	π	$3\pi/2$
D	0	π	2π

Ans: A

Nodes : π (destructive); Antinodes : 0 (constructive)

18. A coil contains M turns of insulated copper wire wound on to a cylindrical iron core of diameter